

Exercise 3.08

Line fitting

```
In [1]: import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
```

```
In [1]: data = np.loadtxt('../data/millikan.txt')
```

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In [2]: data = data.T
```

```
In [3]: data
```

```
Out[3]: array([[5.48740e+14, 6.93100e+14, 7.43070e+14, 8.21930e+14, 9.60740e+14,
1.18400e+15],
[5.30900e-01, 1.08420e+00, 1.27340e+00, 1.65980e+00, 2.19856e+00,
3.10891e+00]])
```

```
In [4]: df = pd.read_csv('../data/millikan.txt', sep=' ', names=['x', 'y'])
```

```
In [5]: df
```

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Out[5]:
```

	x	y
0	5.487400e+14	0.53090
1	6.931000e+14	1.08420
2	7.430700e+14	1.27340
3	8.219300e+14	1.65980
4	9.607400e+14	2.19856
5	1.184000e+15	3.10891

```
In [6]: n = len(df)

ex = 1/n*np.sum(df['x'])
ey = 1/n*np.sum(df['y'])
exx = 1/n*np.sum(df['x']**2)
exy = 1/n*np.sum(df['x']*df['y'])

m = (exy - ex*ey)/(exx - ex**2)
c = (exx*ey - ex*exy)/(exx - ex**2)

print('slope = ', m)
print('y intercept = ', c)

df['fit'] = m*df['x'] + c
```

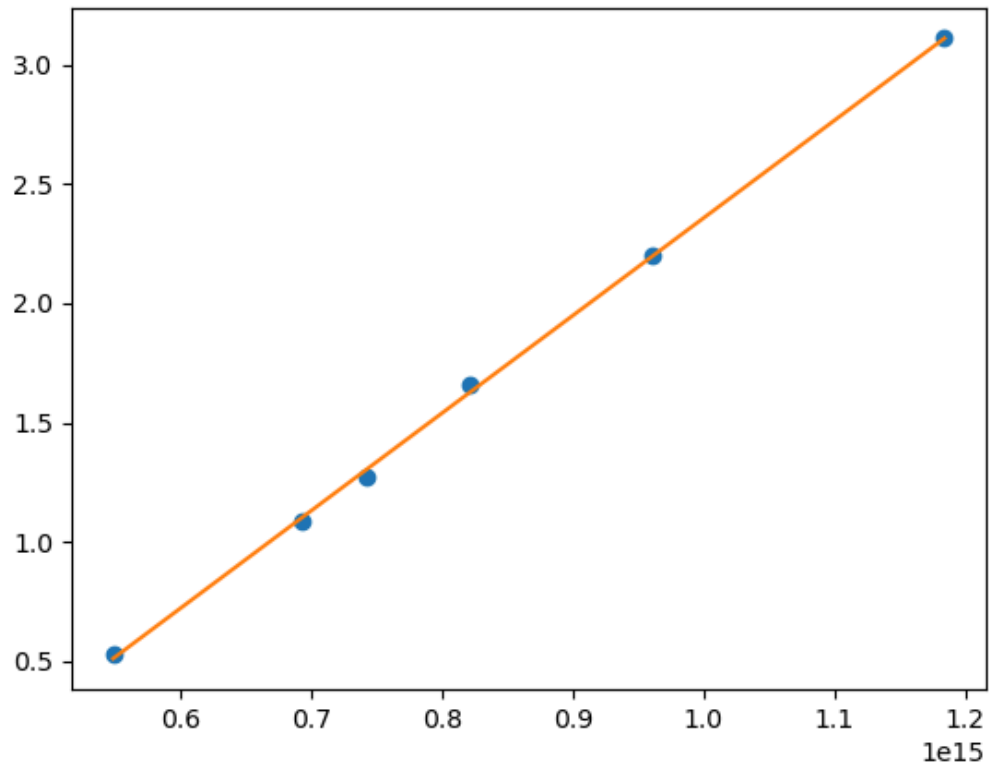
```
fig0, ax0 = plt.subplots()

ax0.plot(df['x'], df['y'], 'o')
ax0.plot(df['x'], df['fit'])
```

```
slope = 4.088227358517502e-15
y intercept = -1.73123580398135
```

Out[6]: [

Figure



In [7]: $m \cdot 1.602 \cdot 10^{(-19)}$

Out[7]: 6.549340228345038e-34

In []: